



ISO/TC 12 "Quantities and units"
Secretariat: SIS
Committee manager: Radhe Sven Mr



Plenary meeting minutes 2025-05-22

Document type	Related content	Document date	Expected action
Meeting / Minutes	Meeting: VIRTUAL 22 May 2025	2025-08-05	INFO

Description

Please find attached Draft Minutes from ISO/TC 12 Plenary meeting on May 22nd.

ISO/TC 12, *Quantities and units*

Secretariat: SIS

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Report of the 22nd Meeting of ISO/TC 12 held on 2025-05-22 virtually via Zoom

1 Opening of the meeting (2025-05-22, 13:00 CEST)

2 Welcome from the host

Prof Göran Grimvall, serving as Acting Chair, welcomed everyone to the meeting. Dr Torgny Carlsson sends his apologies for not being able to attend the meeting.

3 Roll call of delegates

Delegates representing the following countries attended: Austria, China, Finland, Germany, Japan, Portugal, Russian Federation, and Sweden.

See attendance list, [N 1587](#).

4 Work environment: [Presentation](#) on the ISO Code of Ethics and Conduct

5 Adoption of the agenda

The agenda was adopted with the following changes:

- Under item 10: Liaison Reports
 - o Introduction of CalConnect
- Under item 16: AoB
 - o Message from Dr Pellegrino

6 Appointment of the resolution drafting committee

Mr Ronald Tse, Ms Rebecca Cederholm and Ms Anette Lindor Norén were appointed to the resolutions drafting committee.

7 Report of the Committee Manager/Chair and the secretariat

See N1588: [PPT Welcome to the Plenary Meeting of ISO/TC 12, 2025-05-22](#)

See N1581: [Report of the Secretariat to the meeting of ISO/TC 12](#)

Committee Manager, Mr Sven Radhe sent his apologies for not attending the meeting, as his return travel from China had been delayed.

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It was noted that ISO/TC 12 has been in existence since 1947. The scope of ISO/TC 12 includes the standardization of units and symbols for quantities and units, as well as mathematical symbols used across various fields of science and technology. This also involves providing definitions of these quantities and units where necessary, along with standard conversion factors between different units.

As of the date of the meeting, ISO/TC 12 holds direct responsibility for 13 published ISO standards and 12 ISO standards currently under development.

There have been changes in membership since the last ISO/TC 12 plenary meeting. Mauritius has upgraded its status from an Observer (O) member to a Participating (P) member, while Israel has been removed as an O-member.

New liaison representatives were introduced: Dr Torgny Carlsson for CIE, Dr Pellegrino for IEC/TC 25, and Mme Michèle Désenfant for ISO/TC 69.

A new A-liaison with CalConnect was also established, with Mr Michael Douglass, Mr Janssen Gershon, and Mr Ronald Tse appointed as liaison representatives.

The leadership and secretariat for ISO/TC 12 were confirmed as follows: Dr Torgny Carlsson continues as Chair, with his term ending at the end of 2027. Mr Sven Radhe serves as the Committee Manager, supported by Ms Anette Lindor Norén and, newly appointed, Ms Rebecca Cederholm.

8 Strategic issues / Review Strategic Business Plan vers.12

ISO/TC 12 decided to update the SBP, [N 1488](#). After the ISO/TC 12 meeting

NSBs are invited to send their comments to the secretariat before 2025-06-22, and the secretariat will then create an electronic CIB for the review of the Strategic Business Plan (SBP).

9 ISO/TC 12 participation in other activities

9.1 IEC/TC 25/JWG 2 (Joint IEC/TC 25 and ISO/TC 12), Revision and amendment of IEC-related parts of ISO/IEC 80000-series

See [N 1586: IEC/TC 25 Liaison Report](#) by Dr Pellegrino

The IEC/TC 25 Plenary hybrid meeting was held in Milan on 30 October 2024. The agenda, minutes, and decision list from the meeting are available on the IEC website.

Key developments from the meeting include updates from IEC/TC 25 JWG 2 and the appointment of Ms Daniela Zambelli as the new secretary. Additionally, the revised edition of IEC 80000-13 ED2: Quantities and units – Part 13: Information science and technology has been published.

9.2 JCGM, Joint Committee for Guides in Metrology

Dr Carlsson, registered member in JCGM/WG GUM

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See [N 1579: JCGM GUM Liaison report Plenary meeting May 2025](#)

One of the key focus areas was the ongoing work by the **JCGM Working Group 1 (JCGM-WG1:GUM)** on the **Guide to the Expression of Uncertainty in Measurement (GUM)**.

GUM have been restructured into a modular series of parts, aimed at improving clarity and usability. The proposed new structure includes:

- **Part 1:** Introduction (revised edition of JCGM 104)
- **Part 2:** Concepts (formerly JCGM 105)
- **Part 3:** The current 'grandfathered' GUM (formerly JCGM 100)
- **Part 4:** Role of measurement uncertainty in conformity assessment (formerly JCGM 106)
- **Part 5:** Examples of uncertainty evaluation (formerly JCGM 110)
- **Part 6:** Developing and using measurement models (formerly JCGM 103)
- **Part 7:** Propagation of distributions using a Monte Carlo method (formerly JCGM 101)
- **Part 8:** Extension to any number of output quantities (formerly JCGM 102)
- **Part 9:** Statistical models and data analysis for interlaboratory studies (formerly JCGM 109)
- **Part 10:** Least squares methods (formerly JCGM 107)
- **Part 11:** Bayesian methods (formerly JCGM 108)
- **Part 12:** Basic method for uncertainty propagation

9.2.1 GUM, Guide to the Expression of Uncertainty in Measurement

A JCGM Webinar on "Measurement Uncertainty", scheduled for July 2, 2025. This webinar aims to inform and engage stakeholders on current developments in the field.

9.2.2 VIM, International Vocabulary of Metrology

In addition, the VIM4 Committee Draft 3 (3CD) is expected to be circulated for comments in the third quarter of 2024. This follows extensive efforts to build consensus between the JCGM WG1 and WG2 communities on the concept of "measurement uncertainty" and its associated definitions".

9.3 IEC SyC SM/OF 1, Smart Manufacturing Standards Map

Due to absence of Mr Radhe, no report was presented.

10 Liaison reports

10.1.1 ISO/IEC JTC1

- [ISO/IEC DIS 21838-2, Information technology – Top-level ontologies \(TLO\) – Part 2 : Basic Formal Ontology \(BFO\)](#)

top-level ontology (TLO) =def. ontology that is created to represent the categories that are shared across a maximally broad range of domains

Quantities & units in informatics

CalConnect's TC UNITS manages the UnitsML standard used for encoding and representing units and quantities information in XML.

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Originally developed at NIST and OASIS, UnitsML has now its formal home in CalConnect. Since the establishment of TC UNITS, UnitsML has seen several important adoptions, including within the OCX ship-building schemas as the authoritative method to encode units and quantities. Most recently, the BIPM's machine-readable SI Brochures has adopted UnitsML as its encoding and relies on the UnitsML to MathML rendering for presentation. The data set of units and quantities backing UnitsML, called UnitsDB, provides one of the most complete sets of units today.

In addition, TC UNITS is also interested in developing a machine-readable version of the units currently embedded inside the ISO and IEC 80000 publications.

10.1.2 ISO/TC 215**Quantities and Units**

Joint Initiative: ISO/TC 215, ISO/TC 12, IEC/TC 25

Subject: Global Survey of Standards for Quantities and Units in E-Health

A global survey was conducted to assess the current use, gaps, and needs related to quantities and units in electronic health (e-health) systems. Key findings include:

- Widespread variation in the application of units and formats in clinical information systems, particularly in lab data, diagnostic imaging, and drug dosing.
- Limited adoption of international standards such as ISO 80000, VIM, and IEC 80000 series within national e-health frameworks.
- Persistent issues with interoperability, due to inconsistent encoding or absence of reference units in shared health data.
- Need for clearer guidance on the harmonization of SI units in digital health records, decision support systems, and data exchange standards such as HL7 and FHIR.
- Stakeholders reported confusion over definitions (e.g., concentration units, body surface area, metabolic rates) due to legacy system dependencies or incomplete mappings to standard terminologies like SNOMED CT and LOINC.

10.2.3 ISO/TC 43

See N1582, [ISO/TC 43 Liaison Report May 2025](#)

ISO/TC 43 and its subcommittees met in Berlin from 21 to 25 October 2024. During the meeting, a brief report was presented on the current activities of ISO/TC 12 and IEC/TC 25. The next meeting of ISO/TC 43 and its subcommittees is scheduled to take place in Adelaide, Australia, in August 2026.

Comments were provided on the Committee Draft (CD) of JCGM GUM-5 in December 2024. It was noted that the draft lacked examples of uncertainty calculations for data expressed in levels (e.g., decibels), which require conversion to ratios or percentages. Including such examples would be helpful to ensure consistency among practitioners when handling uncertainty calculations for level data. IEC/TC 29 – Electroacoustics recently published a white paper describing this process and the magnitude of the

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overestimation error that can occur if uncertainty calculations are performed directly in decibels. The white paper is available [here](#). It was recommended that this paper be included as a bibliographic reference in the next edition of GUM-5.

ISO 80000 – Part 8: 2020 – Acoustics is currently under systematic review. No errors or major additions have been identified so far. Unless national committees indicate otherwise, the informal recommendation is to confirm the standard without changes.

The next Committee Draft of ISO 80000-15, *Quantities and units – Part 15: Logarithmic quantities and their units*, is apparently pending. The committee looks forward to receiving this draft for review.

10.2.4 Presentation of CalConnect

See N1585, [PPT Introducing CalConnect](#) and N1584, [Introducing UnitsML](#)

CalConnect is an international consortium dedicated to advancing calendaring and scheduling interoperability across the internet. It brings together a diverse, global membership comprising technology companies, calendar and scheduling service providers, software developers, standards organizations, and industry experts.

International Scope

CalConnect operates worldwide, with members spanning multiple continents. Members include major technology companies, software vendors, cloud service providers, and standards bodies. Participation ranges from large multinational corporations to smaller innovative developers.

ISO Liaisons

CalConnect maintains official liaison relationships with several ISO technical committees relevant to quantities, units, and information technology standards.

CalConnects liaisons with ISO:

- ISO/TC 20/SC 6 Standard atmosphere
- ISO/TC 37/ SC 4 Language resource management
- ISO/TC 37/SC 5 Translation, interpreting and related technology
- ISO/TC 46/SC 4 Technical interoperability
- ISO/TC 46/SC 9 Identification and description
- ISO/TC 154 Processes, data elements and documents in commerce, industry and administration
- ISO/TC 211 Geographic information/Geomatics
- ISO/IEC JTC 1/SC 27 Information security, cybersecurity and privacy protection

11 Review of liaisons

Agenda item 11, due to the lack of a Liaison Representative, ISO/TC 12 will create a CIB to see if there is any interest in becoming the liaison representative in ISO/TC 85 Nuclear energy, nuclear technologies, and radiological protection. If there is no interest, we will remove the liaison.

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12 Status of all items of the portfolio and actions to be taken

Current work programme:

- ISO 80000-3 – 5 and 7 – 12 *Quantities and units* – *Amendments*
To address all comments related to the amendments of ISO 80000 parts 3, 5, and 7 through 12, ISO/TC 12 will establish a dedicated Task Force to coordinate the review and resolution process.

A meeting will be called in the fall to discuss the outcomes of the Task Force.

Update on target dates for work in progress:

Standard and/or project under the direct responsibility of ISO/TC 12 Secretariat	Stage
IEC/CD 80000-17 , Quantities and units — Part 17: Time dependency	30.00
IEC/CD 80000-15 , Quantities and units — Part 15: Logarithmic and related quantities	30.60
ISO 80000-5:2019/Amd 1 , Quantities and units — Part 5: Thermodynamics — Amendment 1	60.00
ISO 80000-7:2019/Amd 1 , Quantities and units — Part 7: Light and radiation — Amendment 1	60.00
ISO 80000-8:2020/Amd 1 , Quantities and units — Part 8: Acoustics — Amendment 1	60.00
ISO 80000-9:2019/PRF Amd 1 , Quantities and units — Part 9: Physical chemistry and molecular physics — Amendment 1	50.00
ISO 80000-10:2019/Amd 1 , Quantities and units — Part 10: Atomic and nuclear physics — Amendment 1	60.00
ISO 80000-11:2019/Amd 1 , Quantities and units — Part 11: Characteristic numbers — Amendment 1	60.00
ISO 80000-12:2019/Amd 1 , Quantities and units — Part 12: Condensed matter physics — Amendment 1	60.00
IEC/CD 80000-15 , Quantities and units — Part 15: Logarithmic and related quantities	30.60
IEC/CD 80000-16 , Quantities and units — Part 16: Printing and writing rules	30.60
IEC/CD 80000-17 , Quantities and units — Part 17: Time dependency	30.00

Systematic review results:

Reference	Date closed	Result
ISO 80000-8:2020 (Ed 2), Quantities and units — Part 8: Acoustics	2025-06-05	Pending
ISO 80000-11:2019 (Ed 2), Quantities and units — Part 11: Characteristic numbers	2025-03-05	Confirm
ISO 80000-2:2019 (Ed 2), Quantities and units — Part 2: Mathematics	2024-12-03	Confirm
ISO 80000-4:2019 (Ed 2), Quantities and units — Part 4: Mechanics	2024-12-03	Confirm
ISO 80000-5:2019 (Ed 2), Quantities and units — Part 5: Thermodynamics	2024-12-03	Confirm
ISO 80000-7:2019 (Ed 2), Quantities and units — Part 7: Light and radiation	2024-12-03	Confirm
ISO 80000-9:2019 (Ed 2), Quantities and units — Part 9: Physical chemistry and molecular physics	2024-12-03	Confirm
ISO 80000-10:2019 (Ed 2), Quantities and units — Part 10: Atomic and nuclear physics	2024-12-03	Confirm
ISO 80000-12:2019 (Ed 2), Quantities and units — Part 12: Condensed matter physics	2024-07-15	Confirm

ISO/TC 12, *Quantities and units***13 Items for future work**

Based on comments received during the systematic review stage from CalConnect and other members, ISO/TC 12 decided to initiate a revision of ISO 80000-2:2019.

Rounding Rules – Annex B, part 1

Dr Pellegrino raised the issue of the "rounding matter" referenced in Annex B of part 1, noting that ISO/TC 69 has developed a dedicated standard on rounding rules. He questioned the continued relevance of Annex B if TC 69 proceeds with its new standard.

Dr Pellegrino reiterated that if TC 69 proceeds with a specific standard on rounding, the role and purpose of Annex B should be reconsidered.

The decision from ISO/TC 69 was made not to accept the proposal and to retain the current general rounding rules as described in Annex B. Dr Knopf emphasized the importance of maintaining general guidance rather than adopting specific rules.

Prof Grimvall concluded that **no final decision was taken during this meeting.**

14 Communication (Press releases on publication, etc.)**Promotion of the New Binary Prefix Standard**

Dr Pellegrino proposed that ISO/TC 12 issue a communication to promote the newly published standard on binary prefixes, IEC 80000-13. Although the standard was released earlier this year, it has yet to be included in the SI Brochure. Nevertheless, it presents a valuable opportunity to highlight both the standard itself and the broader work of the committee.

This new development builds on Resolution 3 of the 27th Meeting of the General Conference on Weights and Measures (CGPM), which expanded the range of SI prefixes to cover values from 10^{30} to 10^{-30} . The newly introduced prefixes—quetta, ronna, ronto, and quecto—have now been mirrored in the binary system through the updated IEC 80000-13:2025 standard, *"Quantities and Units – Part 13: Information Science and Technology."*

Using a consistent naming convention, the new binary prefixes were formed by combining the first two letters of the corresponding SI prefix with the syllable "bi", derived from "binary." As a result, robi (symbol Ri, for 2^{90}) and quebi (symbol Qi, for 2^{100}) have been introduced. This extends the binary scale in a way that aligns with earlier prefixes such as kibi (Ki), mebi (Mi), and gibi (Gi).

Because SI prefixes represent powers of 10 and binary prefixes are based on powers of 2, clear distinctions are crucial. This is especially important in fields like data science and information technology, where the use of bits and bytes is common. For example, the quebi factor is approximately 26.8% larger than the quetta factor, illustrating the potential for confusion if binary and decimal multiples are not clearly differentiated.

In response to a question from Prof Grimvall about potential recipients for a press release, Dr Pellegrino recommended contacting ISO Communications, who would be well placed to advise on effective dissemination. He also emphasized that this initiative could serve as a valuable opportunity to raise the profile of ISO/TC 12 and its ongoing contributions to

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clarity and precision in scientific communication.

15 Next ISO/TC 12-meeting 2026

ISO/TC 12 decided on the following meeting arrangements:

- **2025:** Mr Radhe will be asked to circulate a poll proposing dates for a meeting at which the **Task Force** will report back to ISO/TC 12.
- **2026:** A plenary meeting will be held **on 26 May 2026 via Zoom (hybrid)**.

If any member country is interested in hosting the 2026 meeting, they are invited to express their interest. In that case, the meeting will be held in **hybrid format** to enable both in-person and remote participation.

16 Any other business

Dr Pellegrino received an interesting e-mail message from the interim convenor of IEC/ISO/JTC 3/WG 9 (Quantities and terminology), Mr John Devaney, with work item to develop a standard for general quantities for quantum technologies, based on what has been doing in CEN/CENELEC/JTC 22/WG 2 (Metrology, sensors, enabling tech), with Mr John Devaney as convenor.

Dr Pellegrino will contact CEN/CENELEC/JTC 22/WG 2 to ask them what expertise their needs are.

17 Approval of resolutions

See, N1583, [Resolutions taken at the Plenary meeting 2025-05-22](#)

18 Closure of the meeting 2025-05-22, (16:00 CEST)

Prof Grimvall thanked the delegates for their participation and closed the meeting.